Applied ML Report

1. Introduction

This project implements a Healthcare Evidence-Based Assistant using Retrieval-Augmented Generation (RAG). The assistant answers healthcare-related questions by retrieving relevant information from WHO COVID-19 Guidelines and generating responses based only on the retrieved evidence. This approach helps reduce hallucinations and provides more reliable answers.

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2. Problem Statement

The objective of this project is to build a healthcare assistant capable of answering user questions using trusted medical documents. The assistant retrieves relevant information from the WHO COVID-19 Guidelines and generates evidence-based responses along with the retrieved source context.

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3. Dataset

The project uses the WHO COVID-19 Guidelines PDF as the knowledge base.

Document Details:

- Source: World Health Organization (WHO)
- Number of Pages: 58
- Total Characters Extracted: 165,730
- Number of Text Chunks: 415

The document was converted into text and divided into overlapping chunks to improve retrieval quality.

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4. System Architecture

The system follows a Retrieval-Augmented Generation (RAG) pipeline.

Workflow:

1. User enters a healthcare-related question.
2. The question is converted into an embedding using Sentence Transformers.
3. FAISS retrieves the most relevant text chunks from the WHO document.
4. The retrieved context is passed to the FLAN-T5 language model.
5. The model generates an answer based on the retrieved evidence.
6. The answer is displayed through a Gradio web interface.

5. Methodology

The implementation consists of the following stages:

• PDF text extraction using PyMuPDF.

• Text chunking with overlapping windows.

• Sentence embedding generation using all-MiniLM-L6-v2.

• Vector database creation using FAISS.

• Semantic retrieval of relevant document chunks.

• Answer generation using Google's FLAN-T5 model.

• Deployment through a Gradio web interface.

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6. Technologies Used

Programming Language:

- Python

Libraries:

- PyTorch
- Sentence Transformers
- FAISS
- Transformers
- PyMuPDF
- NumPy
- Gradio

Development Environment:

- Google Colab

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7. Evaluation

The system was evaluated using manual testing.

Example questions included:

- What are the symptoms of COVID-19?
- How does COVID-19 spread?
- How can COVID-19 be prevented?
- Who is at higher risk?

The assistant successfully retrieved relevant information from the WHO guidelines and generated context-based answers.

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8. Example Queries

Example 1: Question: What are the symptoms of COVID-19?

Result: The assistant retrieved the relevant section from the WHO guidelines and generated an evidence-based response.

Example 2: Question: How does COVID-19 spread?

Result: The assistant returned information retrieved from the WHO document describing transmission methods.

Example 3: Question: How can COVID-19 be prevented?

Result: The assistant generated preventive recommendations based on the retrieved WHO guidance.

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9. Limitations

- The system is limited to the uploaded WHO document.
- Responses depend on the quality of retrieved chunks.
- The assistant does not replace professional medical advice.
- Performance may vary for questions not covered in the document.

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10. Future Improvements

- Support multiple medical documents.
- Improve confidence estimation.
- Add conversation memory.
- Include multilingual support.
- Integrate more advanced medical language models.
- Deploy the application as a cloud-based web service.

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11. Conclusion

This project demonstrates a Retrieval-Augmented Generation based Healthcare Assistant capable of retrieving relevant information from WHO COVID-19 Guidelines and generating evidence-based responses. The combination of semantic search using FAISS and answer generation using FLAN-T5 provides an effective solution for healthcare question answering while reducing hallucinations by grounding responses in trusted medical documents.